Read in catalog information from a text file and plot some parameters

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Learning Goals

- Read an ASCII file using `astropy.io`
- Convert between representations of coordinate components using `astropy.coordinates` (hours to degrees)
- Make a spherical projection sky plot using `matplotlib`

Keywords

file input/output, coordinates, tables, units, scatter plots, matplotlib

Summary

This tutorial demonstrates the use of `astropy.io.ascii` for reading ASCII data, `astropy.coordinates` and `astropy.units` for converting RA (as a sexagesimal angle) to decimal degrees, and `matplotlib` for making a color-magnitude diagram and on-sky locations in a Mollweide projection.

```
import numpy as np

# Set up matplotlib
import matplotlib.pyplot as plt

%matplotlib inline
```

Astropy provides functionality for reading in and manipulating tabular data through the `astropy.table` subpackage. An additional set of tools for reading and writing ASCII data are provided with the `astropy.io.ascii` subpackage, but fundamentally use the classes and methods implemented in `astropy.table`.

We'll start by importing the `ascii` subpackage:

```
from astropy.io import ascii
```

For many cases, it is sufficient to use the `ascii.read('filename')` function as a black box for reading data from table-formatted text files. By default, this function will try to figure out how your data is formatted/delimited (by default, `guess=True`). For example, if your data are:

```
# name,ra,dec
BLG100,17:51:00.0,-29:59:48
BLG101,17:53:40.2,-29:49:52
BLG102,17:56:20.2,-29:30:51
BLG103,17:56:20.2,-30:06:22
...
```

(see *simple_table.csv*)

`ascii.read()` will return a `Table` object:

```
tbl = ascii.read("simple_table.csv")
tbl
```

Table length=4

name	ra	dec
str6	str10	str9
BLG100	17:51:00.0	-29:59:48
BLG101	17:53:40.2	-29:49:52
BLG102	17:56:20.2	-29:30:51
BLG103	17:56:20.2	-30:06:22

The header names are automatically parsed from the top of the file, and the delimiter is inferred from the rest of the file – awesome! We can access the columns directly from their names as 'keys' of the table object:

```
tbl["ra"]
```

<Column name='ra' dtype='str10' length=4>

17:51:00.0

17:53:40.2

17:56:20.2

17:56:20.2

If we want to then convert the first RA (as a sexagesimal angle) to decimal degrees, for example, we can pluck out the first (0th) item in the column and use the `coordinates` subpackage to parse the string:

```
import astropy.coordinates as coord
import astropy.units as u

first_row = tbl[0]  # get the first (0th) row
ra = coord.Angle(first_row["ra"], unit=u.hour)  # create an Angle object
ra.degree  # convert to degrees
```

```
np.float64(267.75)
```

Now let's look at a case where this breaks, and we have to specify some more options to the `read()` function. Our data may look a bit messier::

```
,,,,2MASS Photometry,,,,,,WISE Photometry,,,,,,,,Spectra,,,,Astrometry,,,,,,,,,,,
Name,Designation,RA,Dec,Jmag,J_unc,Hmag,H_unc,Kmag,K_unc,W1,W1_unc,W2,W2_unc,W3,W3_u
,00 04 02.84 -64 10 35.6,1.01201,-64.18,15.79,0.07,14.83,0.07,14.01,0.05,13.37,0.03,
PC 0025+04,00 27 41.97 +05 03 41.7,6.92489,5.06,16.19,0.09,15.29,0.10,14.96,0.12,14.
...
```

(see *Young-Objects-Compilation.csv*)

If we try to just use `ascii.read()` on this data, it fails to parse the names out and the column names become `col` followed by the number of the column:

```
tbl = ascii.read("Young-Objects-Compilation.csv")
tbl.colnames
```

```
['col1',
 'col2',
 'col3',
 'col4',
 'col5',
 'col6',
 'col7',
 'col8',
 'col9',
 'col10',
 'col11',
 'col12',
 'col13',
 'col14',
 'col15',
 'col16',
 'col17',
 'col18',
 'col19',
 'col20',
 'col21',
 'col22',
 'col23',
 'col24',
 'col25',
 'col26',
 'col27',
 'col28',
 'col29',
 'col30',
 'col31',
 'col32',
 'col33',
 'col34']
```

What happened? The column names are just `col1`, `col2`, etc., the default names if `ascii.read()` is unable to parse out column names. We know it failed to read the column names, but also notice that the first row of data are strings – something else went wrong!

```
tbl[0]
```

Row index=0

col1	col2	col3	col4	col5	col6	col7	col8	col9	col10	col11	col12	c
str24	**str25**	**str9**	**str6**	**str16**	**str5**	**str5**	**str5**	**str5**	**str5**	**str15**	**str6**	
--	--	--	--	2MASS Photometry	--	--	--	--	--	WISE Photometry	--	

A few things are causing problems here. First, there are two header lines in the file and the header lines are not denoted by comment characters. The first line is actually some meta data that we don't care about, so we want to skip it. We can get around this problem by specifying the `header_start` keyword to the `ascii.read()` function. This keyword argument specifies the index of the row in the text file to read the column names from:

```
tbl = ascii.read("Young-Objects-Compilation.csv", header_start=1)
tbl.colnames
```

```
['Name',
 'Designation',
 'RA',
 'Dec',
 'Jmag',
 'J_unc',
 'Hmag',
 'H_unc',
 'Kmag',
 'K_unc',
 'W1',
 'W1_unc',
 'W2',
 'W2_unc',
 'W3',
 'W3_unc',
 'W4',
 'W4_unc',
 'Spectral Type',
 'Spectra (FITS)',
 'Opt Spec Refs',
 'NIR Spec Refs',
 'pm_ra (mas)',
 'pm_ra_unc',
 'pm_dec (mas)',
 'pm_dec_unc',
 'pi (mas)',
 'pi_unc',
 'radial velocity (km/s)',
 'rv_unc',
 'Astrometry Refs',
 'Discovery Refs',
 'Group/Age',
 'Note']
```

Great! Now the columns have the correct names, but there is still a problem: all of the columns have string data types, and the column names are still included as a row in the table. This is because by

default the data are assumed to start on the second row (index=1). We can specify `data_start=2` to tell the reader that the data in this file actually start on the 3rd (index=2) row:

```
tbl = ascii.read("Young-Objects-Compilation.csv", header_start=1, data_start=2)
```

Some of the columns have missing data, for example, some of the `RA` values are missing (denoted by – when printed):

```
print(tbl["RA"])
```

```
    RA
---------
  1.01201
  6.92489
  8.23267
  9.42942
 11.33929
       --
       --
       --
 21.19163
  21.5275
      ...
       --
303.46467
   321.71
       --
       --
332.05679
333.43715
342.47273
       --
350.72079
Length = 64 rows
```

This is called a **Masked column** because some missing values are masked out upon display. If we want to use this numeric data, we have to tell `astropy` what to fill the missing values with. We can do this with the `.filled()` method. For example, to fill all of the missing values with `NaN`'s:

```
tbl["RA"].filled(np.nan)
```

```
<Column name='RA' dtype='float64' length=64>
                 1.01201
                 6.92489
                 8.23267
                 9.42942
                11.33929
                     nan
                     nan
                     nan
                21.19163
                 21.5275
                25.49263
                     nan
                     ...
               300.20171
                     nan
               303.46467
                  321.71
                     nan
                     nan
               332.05679
               333.43715
               342.47273
                     nan
               350.72079
```

Let’s recap what we’ve done so far, then make some plots with the data. Our data file has an extra line above the column names, so we use the `header_start` keyword to tell it to start from line 1

instead of line 0 (remember Python is 0-indexed!). We then used had to specify that the data starts on line 2 using the `data_start` keyword. Finally, we note some columns have missing values.

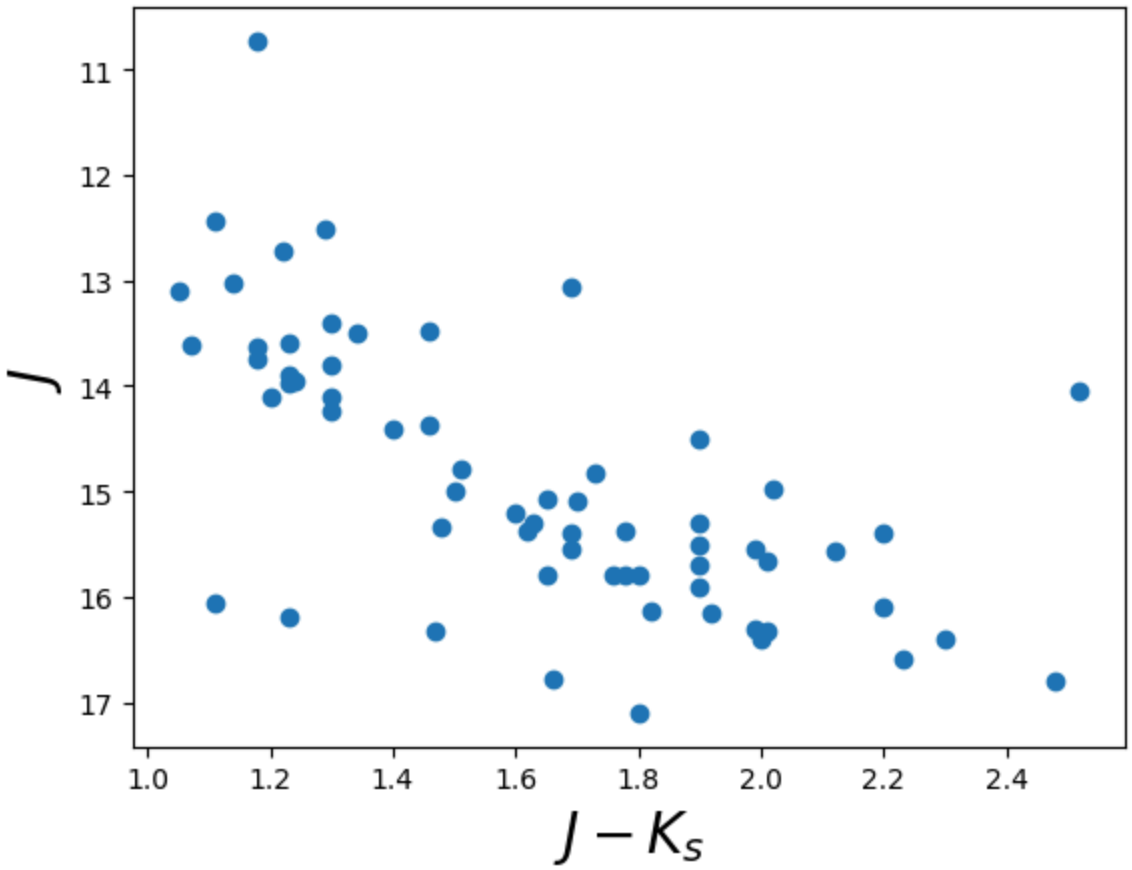
```
data = ascii.read("Young-Objects-Compilation.csv", header_start=1, data_start=2)
```

Now that we have our data loaded, let's plot a color-magnitude diagram.

Here we simply make a scatter plot of the J-K color on the x-axis against the J magnitude on the y-axis. We use a trick to flip the y-axis `plt.ylim(reversed(plt.ylim()))`. Called with no arguments, `plt.ylim()` will return a tuple with the axis bounds, e.g. (0,10). Calling the function *with* arguments will set the limits of the axis, so we simply set the limits to be the reverse of whatever they were before. Using this `matplotlib`-style plotting is convenient for making quick plots and interactive use, but is not great if you need more control over your figures.

```
plt.scatter(data["Jmag"] - data["Kmag"], data["Jmag"])  # plot J-K vs. J
plt.ylim(reversed(plt.ylim()))  # flip the y-axis
plt.xlabel("$J-K_s$", fontsize=20)
plt.ylabel("$J$", fontsize=20)
```

```
Text(0, 0.5, '$J$')
```



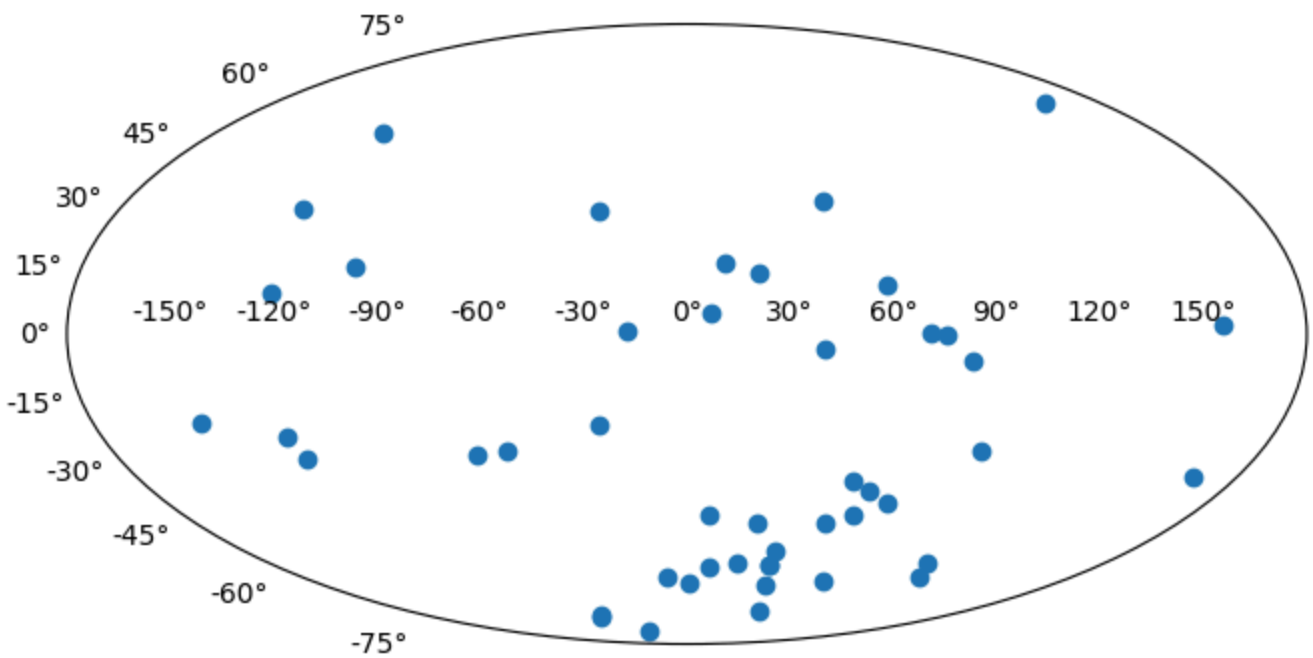
As a final example, we will plot the angular positions from the catalog on a 2D projection of the sky. Instead of using `matplotlib`-style plotting, we'll take a more object-oriented approach. We'll start by creating a `Figure` object and adding a single subplot to the figure. We can specify a projection with the `projection` keyword; in this example we will use a Mollweide projection. Unfortunately, it is highly non-trivial to make the `matplotlib` projection defined this way follow the celestial convention of longitude/RA increasing to the left.

The axis object, `ax`, knows to expect angular coordinate values. An important fact is that it expects the values to be in *radians*, and it expects the azimuthal angle values to be between (-180º,180º). This is (currently) not customizable, so we have to coerce our RA data to conform to these rules! `astropy` provides a coordinate class for handling angular values, `astropy.coordinates.Angle`. We can convert our column of RA values to radians, and wrap the angle bounds using this class.

```
ra = coord.Angle(data["RA"].filled(np.nan) * u.degree)
ra = ra.wrap_at(180 * u.degree)
dec = coord.Angle(data["Dec"].filled(np.nan) * u.degree)
```

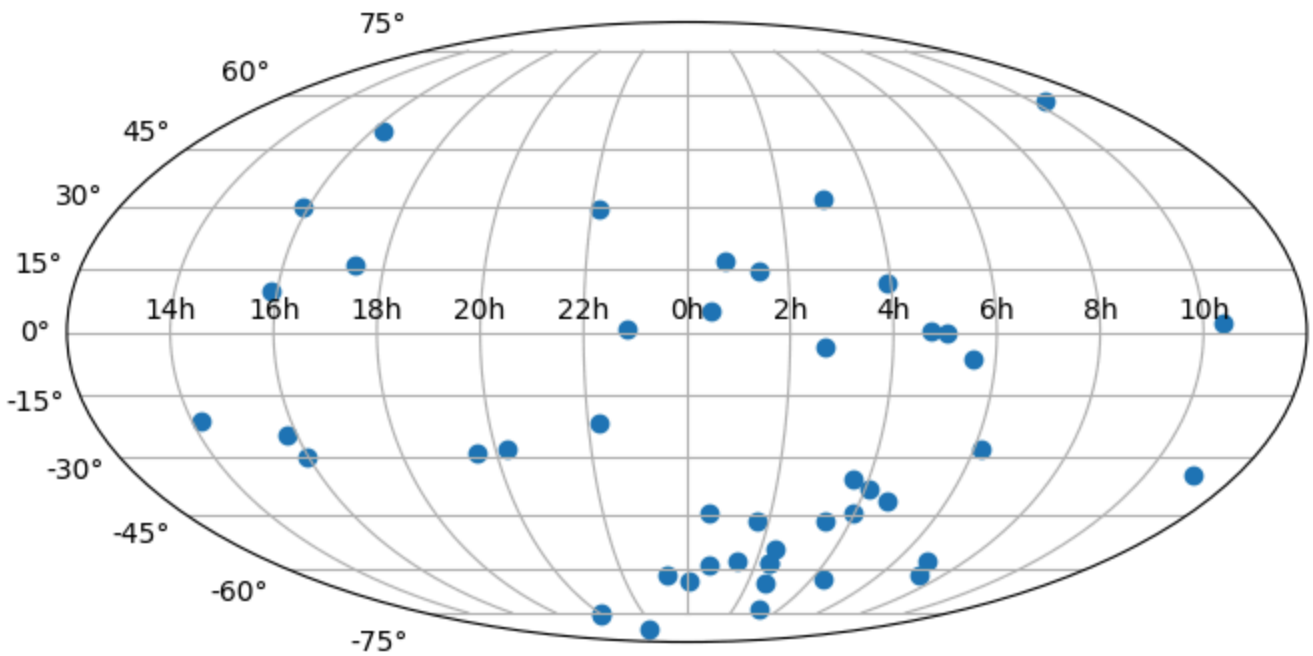
```
fig = plt.figure(figsize=(8, 6))
ax = fig.add_subplot(111, projection="mollweide")
ax.scatter(ra.radian, dec.radian)
```

```
<matplotlib.collections.PathCollection at 0x7f1b54d643b0>
```



By default, matplotlib will add degree tick labels, so let's change the horizontal (x) tick labels to be in units of hours, and display a grid:

```
fig = plt.figure(figsize=(8, 6))
ax = fig.add_subplot(111, projection="mollweide")
ax.scatter(ra.radian, dec.radian)
ax.set_xticklabels(
    ["14h", "16h", "18h", "20h", "22h", "0h", "2h", "4h", "6h", "8h", "10h"]
)
ax.grid(True)
```



We can save this figure as a PDF using the `savefig` function:

```
fig.savefig("map.pdf")
```

Exercises

Make the map figures as just above, but color the points by the `'Kmag'` column of the table.

Try making the maps again, but with each of the following projections: `aitoff`, `hammer`, `lambert`, and `None` (which is the same as not giving any projection). Do any of them make the data seem easier to understand?